

SIM2.2 — 035 NMOS Channel-Length Modulation

Setup and selected bias

This simulation uses the NM Level 8 BSIM3 model and `nmos_035` symbol from the supplied 035 LTspice library. The source and bulk are grounded. The aspect ratio is held at $W/L = 10$ for all five devices.

The assignment requires a constant overdrive but does not prescribe its numerical value or the drain-voltage sweep endpoints. We choose a nominal $VOV = 0.500$ V as a clear, practical bias point. For the supplied model, the threshold at the 1 V reference point is about 0.500–0.536 V across the five lengths, so this target gives V_{GS} values near 1.0 V. V_{GS} is calibrated separately for each length from the LTspice operating-point V_{th} at $V_{DS} = 1.0$ V, using $V_{GS} = V_{th} + VOV$; this keeps the nominal overdrive comparable as L changes.

L (μm)	W (μm)	Vth at VDS = 1 V (V)	VGS (V)	Nominal VOV (V)
1	10	0.536	1.036	0.500
2	20	0.518	1.018	0.500
5	50	0.506	1.006	0.500
10	100	0.502	1.002	0.500
20	200	0.500	1.000	0.500

The full characteristic is swept from $V_{DS} = 0$ V to 2.0 V in 10 mV steps. This displays the low- V_{DS} triode region as well as the saturation region. The upper endpoint of 2.0 V is a selected sweep limit that gives a broad drain-voltage range for observing the output slope.

With the simple long-channel criterion, the saturation boundary is $V_{DS} \approx V_{GS} - V_{th} = VOV$, nominally about 0.5 V here. We begin the λ calculation at 0.6 V, 0.1 V above that nominal boundary, and use the saturation interval from 0.6 V to 2.0 V. The choice of 0.6 V provides margin from the knee; points below it remain in the full-characteristic plot but are excluded from the saturation fit. These numeric limits are chosen for this simulation, not specified by the assignment.

The gate voltages are fixed during each drain sweep. Because the BSIM3 model can shift its effective V_{th} slightly with V_{DS} , the 0.500 V overdrive is matched at the 1.0 V reference point and is nominal over the sweep rather than mathematically exact at every point.

Method

In saturation, use the assignment's approximation:

$$I_D \approx I_{D0} (1 + \lambda V_{DS})$$

Estimate the average slope over the selected saturation interval, using only the simulated values at 0.6 V and 2.0 V:

$$g_{ds} \approx [I_D(2.0 \text{ V}) - I_D(0.6 \text{ V})] / 1.4 \text{ V}; \quad I_{D0} \approx I_D(0.6 \text{ V}) - 0.6 g_{ds}; \quad \lambda \approx g_{ds} / I_{D0}.$$

The assignment does not specify which saturation-region bias point should be used for I_D in the r_o formula. Here we choose $V_{DS} = 1.3$ V, the midpoint of the 0.6–2.0 V interval used for the saturation fit. This keeps the reported resistance away from both fit endpoints and gives one representative operating point for comparing the five devices. A different saturation-region V_{DS} would give a slightly different r_o , because $r_o = 1/(\lambda I_D)$ depends on the bias current.

Then calculate output resistance at the selected $V_{DS} = 1.3$ V using the requested formula:

$$r_o \approx 1 / [\lambda I_D(1.3 \text{ V})].$$

The sweep starts at 0 V for the plot, but samples below 0.6 V are not used in these saturation-region calculations.

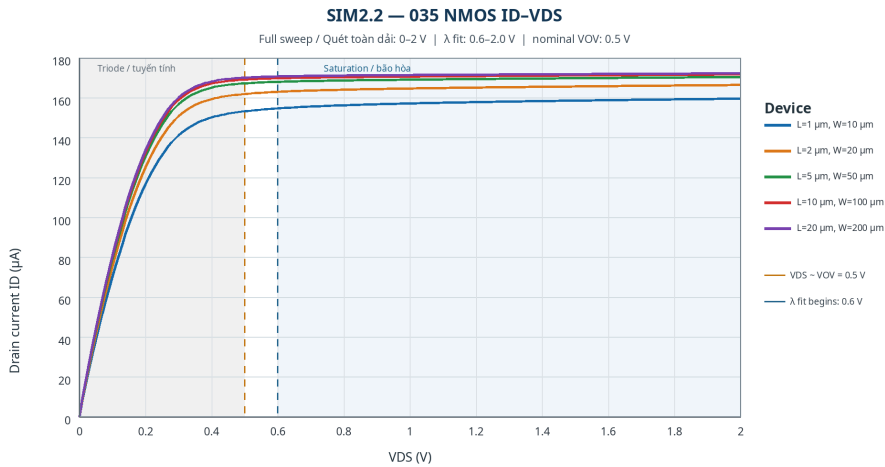
Results

L (μm)	W (μm)	ID at 1.3 V (μA)	Slope gds (μS)	λ (V ⁻¹)	ro (kΩ)
1	10	158.037	3.489	0.022876	276.611
2	20	165.261	2.464	0.015263	396.453
5	50	169.485	1.729	0.010358	569.653
10	100	170.948	1.379	0.008162	716.706
20	200	171.633	1.097	0.006455	902.644

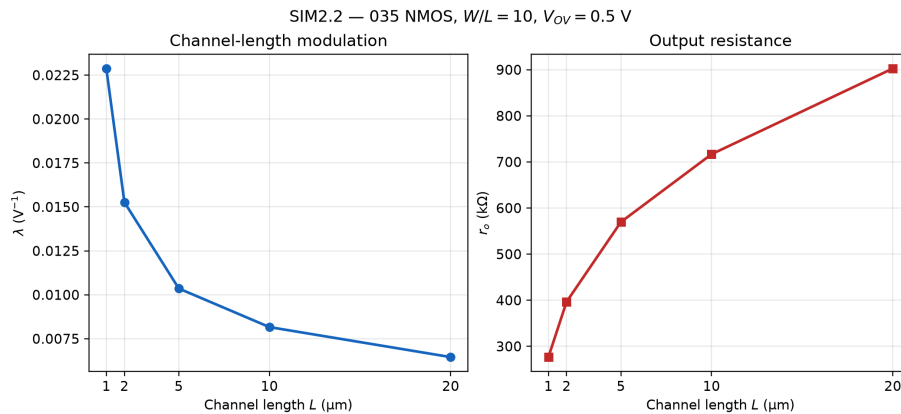
Plots

Full drain-current characteristic, from 0 V through saturation

The shaded low-voltage region shows the triode portion. The blue shading marks the 0.6–2.0 V interval used to estimate λ .



Channel-length modulation and output resistance versus channel length



Discussion

The full I_D - V_{DS} plot shows the nearly linear rise at low V_{DS} , followed by the flatter saturation-region curves. Only the latter interval is used to estimate channel-length modulation, as required by the saturation approximation.

As L increases from $1\ \mu\text{m}$ to $20\ \mu\text{m}$, λ decreases from $0.022876\ \text{V}^{-1}$ to $0.006455\ \text{V}^{-1}$, while r_o increases from $276.6\ \text{k}\Omega$ to $902.6\ \text{k}\Omega$. The output curves become flatter for longer devices, showing weaker channel-length modulation. The current at $1.3\ \text{V}$ rises modestly from $158.0\ \mu\text{A}$ to $171.6\ \mu\text{A}$ even though W/L and nominal overdrive are held constant. This is the behavior of the supplied BSIM3 model across these geometries; the ideal long-channel square-law model would predict less variation.

Files

- `../Simulation/SIM2_2_NMOS.asc` — LTspice schematic; sweeps the full V_{DS} range from 0 to 2.0 V and measures λ and r_o over the saturation interval.
- `../Simulation/SIM2_2_NMOS_VTH.cir` — operating-point threshold calibration at $V_{DS} = 1\ \text{V}$.
- `../Simulation/SIM2_2_NMOS_ID_VDS.csv` — full 0–2 V output-characteristic data.
- `../Simulation/SIM2_2_NMOS_results.csv` — metrics calculated from the 0.6–2.0 V saturation interval.